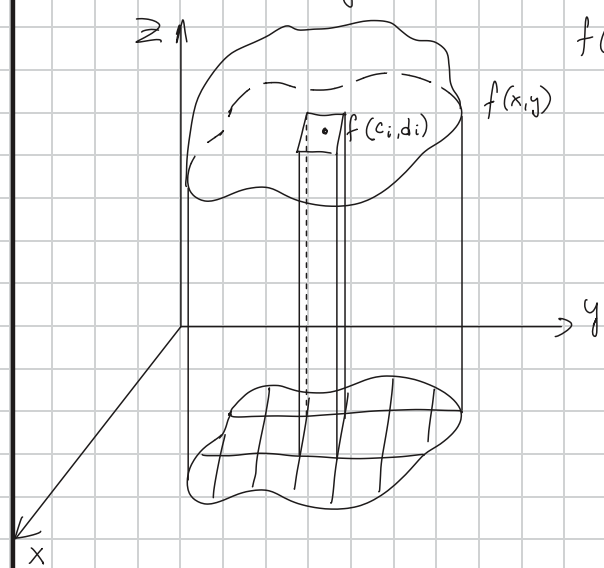


Вычисление двойного интеграла



$$V \approx \sum V_i = \sum \Delta x_i \cdot \Delta y_i \cdot f(c_i, d_i)$$

$$V = \lim \sum f(c_i, d_i) \Delta x_i \Delta y_i$$

Обозначим $\iint_D f(x, y) dx dy$

Формула

$$\iint_D f(x, y) dx dy = \int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x, y) dy \right) dx$$

глобальной *крайней*

Замечание

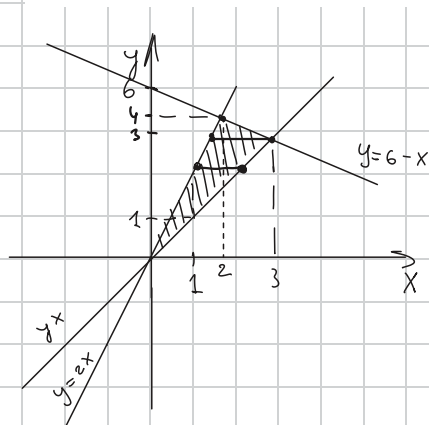
Правый (крайний) интеграл можно без проблем:

$$\int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x, y) dy \right) dx \xrightarrow{\text{замечание}} \int_a^b dx \int_{y_1(x)}^{y_2(x)} f(x, y) dy$$

вычисляем по порядку

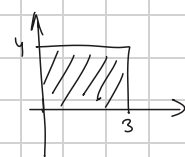
$$\iint_D f(x, y) dx dy \xrightarrow{y=c} \int_c^a dy \int_{x_1(y)}^{x_2(y)} f(x, y) dx$$

3493. Треугольник со сторонами $y = x$, $y = 2x$ и $x + y = 6$.



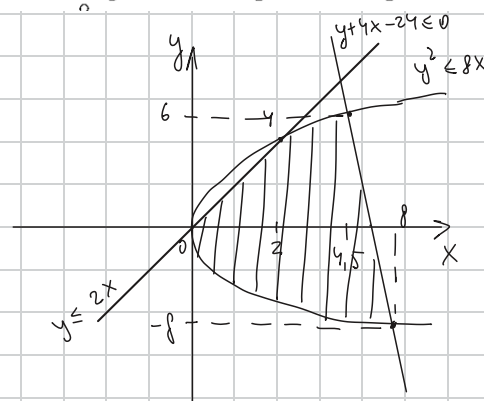
$$\int_0^3 dy \int_{\frac{y}{2}}^y f(x, y) dx$$

$$\int_0^3 dy \int_{\frac{y}{2}}^y f(x, y) dx + \int_3^6 dy \int_{\frac{y}{2}}^{6-y} f(x, y) dx =$$



$$= \int_0^2 dx \int_x^{2x} f(x, y) dy + \int_2^3 dx \int_x^{6-x} f(x, y) dy$$

3496. $y^2 \leq 8x$, $y \leq 2x$, $y + 4x - 24 \leq 0$.



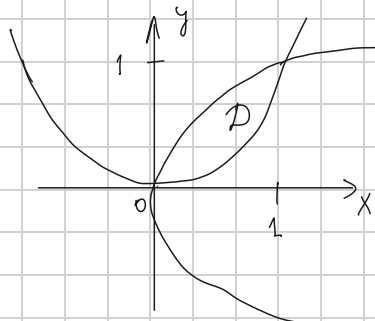
$$\int_0^2 dx \int_{-\sqrt{8x}}^{\sqrt{8x}} f(x, y) dy + \int_2^{4.5} dx \int_{-\sqrt{8x}}^{\sqrt{8x}} f(x, y) dy + \int_{4.5}^8 dx \int_{-\sqrt{8x}}^{4x+24} f(x, y) dy$$

3508. $\iint_D (x^2 + y) dx dy$, D — область, ограниченная параболлами $y = x^2$ и $y^2 = x$.

$$\iint_D (x^2 + y) dx dy = \int_0^1 dx \int_{x^2}^{\sqrt{x}} (x^2 + y) dy$$

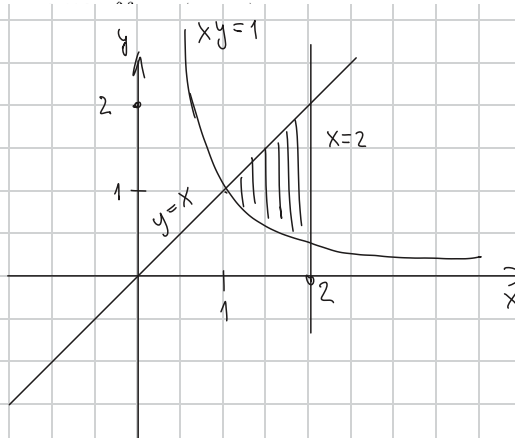
$$\textcircled{1} \int_{x^2}^{\sqrt{x}} (x^2 + y) dy = \int_{x^2}^{\sqrt{x}} x^2 dy + \int_{x^2}^{\sqrt{x}} y dy = x^2 y \Big|_{x^2}^{\sqrt{x}} + \frac{y^2}{2} \Big|_{x^2}^{\sqrt{x}} = x^2 (\sqrt{x} - x^2) + \frac{1}{2} (x - x^4) =$$

$$= x^2 \sqrt{x} - x^4 + \frac{1}{2} x - \frac{1}{2} x^4 = -1.5x^4 + x^2 \sqrt{x} + 0.5x$$



$$\textcircled{2} \left(-1.5 \frac{x^5}{5} \right) \Big|_0^1 + \left(\frac{x^3 \sqrt{x}}{3.5} \right) \Big|_0^1 + \left(0.5 \frac{x^2}{2} \right) \Big|_0^1 = -0.3 + \frac{2}{7} + \frac{1}{4} = \frac{33}{140}$$

3509. $\iint_D \frac{x^2}{y^2} dx dy$, D — область, ограниченная прямыми $x = 2$, $y = x$ и гиперболой $xy = 1$.



$$\int_1^2 dx \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy$$

$$1) \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy = \left[-\frac{x^2}{y} \right]_{\frac{1}{x}}^x =$$

$$= -\frac{x^2}{x} - \left(-x^2 \cdot x \right) = x^3 - x$$

$$2) \int_1^2 (x^3 - x) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_1^2 =$$

$$= \left(\frac{16}{4} - \frac{4}{2} \right) - \left(\frac{1}{4} - \frac{1}{2} \right) = \frac{9}{4}$$

3512. $\iint_D x^2 y^2 \sqrt{1-x^3-y^3} dx dy$, D — область, ограниченная линией $x^3 + y^3 = 1$ и осями координат.

$x=0$
 $y=0$
 $x^3 + y^3 = 1$

$\int_0^1 dx \int_0^{\sqrt[3]{1-x^3}} x^2 y^2 \sqrt{1-x^3-y^3} dy = \frac{1}{3} \int_0^1 dx \left(x^2 \int_0^{\sqrt[3]{1-x^3}} \sqrt{1-x^3-y^3} dy^3 \right)$

$\int_0^{\sqrt[3]{1-x^3}} \sqrt{1-x^3-y^3} d(1-x^3-y^3) = \frac{2}{3} (1-x^3-y^3) \sqrt{1-x^3-y^3} \Big|_0^{\sqrt[3]{1-x^3}} =$

$= \frac{2}{3} (1-x^3 - (1-x^3) \cdot \sqrt{1-x^3-(1-x^3)}) - \frac{2}{3} ((1-x^3) \cdot \sqrt{1-x^3}) =$

$= \frac{2}{3} (1-x^3) \sqrt{1-x^3}$

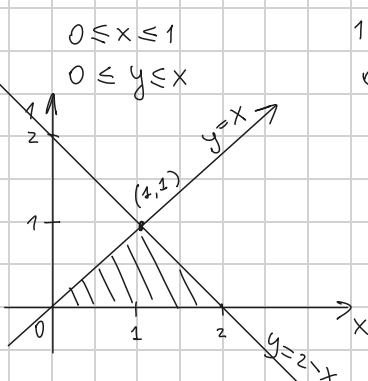
$-\frac{2}{9} \int_0^1 x^2 (1-x^3)^{\frac{3}{2}} dx = \frac{2}{27} \int_0^1 (1-x^3)^{\frac{3}{2}} d(1-x^3) = \frac{2}{27} \cdot \frac{2}{5} (1-x^3)^{\frac{5}{2}} \Big|_0^1 =$

$= \frac{4}{27 \cdot 5} (0-1) = -\frac{4}{27 \cdot 5}$

3504. Переменив порядок интегрирования, записать данное выражение в виде одного двукратного интеграла:

- 1) $\int_0^1 dx \int_0^x f(x, y) dy + \int_1^2 dx \int_0^{2-x} f(x, y) dy;$
- 2) $\int_0^1 dx \int_0^{x^2} f(x, y) dy + \int_1^3 dx \int_0^{(3-x)^2} f(x, y) dy;$
- 3) $\int_0^1 dx \int_0^{x^{\frac{2}{3}}} f(x, y) dy + \int_0^2 dx \int_0^{1-\sqrt{4x-x^2-3}} f(x, y) dy.$

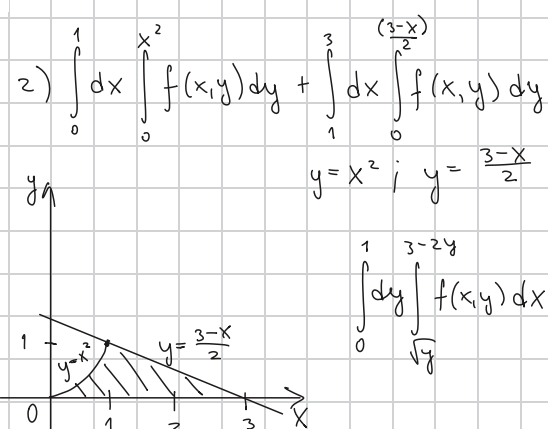
1) $\int_0^1 dx \int_0^x f(x, y) dy + \int_1^2 dx \int_0^{2-x} f(x, y) dy$



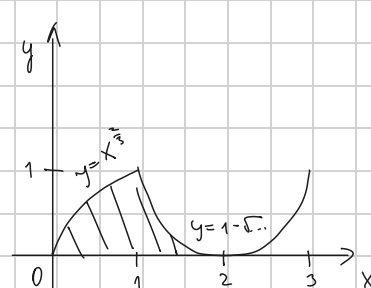
$0 \leq x \leq 1$
 $0 \leq y \leq x$

$1 \leq x \leq 2$
 $0 \leq y \leq 2-x$

$\int_0^1 dy \int_y^{2-y} f(x, y) dx$



3) $\int_0^1 dx \int_0^{x^{\frac{2}{3}}} f(x, y) dy + \int_1^2 dx \int_0^{1-\sqrt{4x-x^2-3}} f(x, y) dy$



$\int_0^1 dy \int_{y^{\frac{3}{2}}}^{2-\sqrt{1-(y-1)^2}} f(x, y) dx$

$(y-1)^2 = (-\sqrt{4x-x^2-3})^2$

$(y-1)^2 = 4x-x^2-3$

$(y-1)^2 - 4x + x^2 + 3 = 0$

$(y-1)^2 + (x-2)^2 = 1$

$(x-2)^2 = 1 - (y-1)^2$

$x-2 = \sqrt{1 - (y-1)^2}$

$x = \sqrt{1 - (y-1)^2} + 2$

3521. $\int_0^{e-1} dx \int_0^{e-x-1} dy \int_e^{x+y+e} \frac{\ln(z-x-y)}{(x-e)(x+y-e)} dz.$

$\int_0^{e-1} dx \int_0^{e-x-1} dy \int_e^{x+y+e} \frac{\ln(z-x-y)}{(x-e)(x+y-e)} dz$

$\int_e^{x+y+e} \frac{\ln(z-x-y)}{(x-e)(x+y-e)} dz = \int_e^{x+y+e} \ln(z-x-y) dz = \frac{1}{(x-e)(x+y-e)} \int_e^{x+y+e} \ln(z-x-y) dz = \frac{1}{(x-e)(x+y-e)} \left((z-x-y) \ln(z-x-y) - (z-x-y) \right) \Big|_e^{x+y+e} =$

$\int_e^{x+y+e} \ln t dt = [u = \ln t, dv = dt] = u \cdot v - \int v du = \ln t \cdot t - \int t \frac{dt}{t} = t \ln t - t + C$

$= \frac{(x+y+e-x-y) \ln(x+y+e-x-y) - (x+y+e-x-y) - (e-x-y) \ln(e-x-y) + (e-x-y)}{(x-e)(x+y-e)}$

3523. $\iiint_{\Omega} xy dx dy dz$, Ω — область, ограниченная гиперболическим параболоидом $z = xy$ и плоскостями $x + y = 1$ и $z = 0$ ($z \geq 0$).

$\iiint_{\Omega} xy dx dy dz$

$$J = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} \iint_{D_{xy}} f(x,y) dx dy = \begin{vmatrix} x=x(u,v) \\ y=y(u,v) \end{vmatrix} = \iint_{D_{uv}} f(x(u,v), y(u,v)) |J| du dv$$

якобиан

Замена в цилиндрические координаты (ρ, φ, z)

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{cases} \quad J = \rho$$

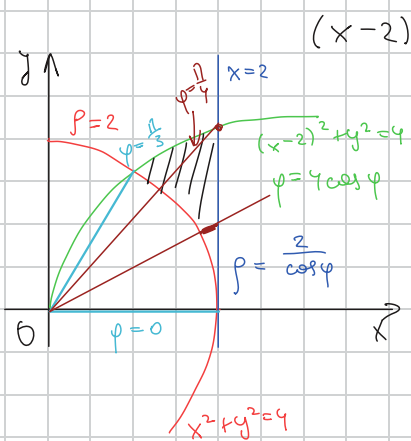
Замена в полярные координаты (ρ, φ)

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{cases} \quad J = \begin{vmatrix} x'_\rho & x'_\varphi \\ y'_\rho & y'_\varphi \end{vmatrix} = \begin{vmatrix} \cos \varphi & -\rho \sin \varphi \\ \sin \varphi & \rho \cos \varphi \end{vmatrix} = \rho \cos^2 \varphi + \rho \sin^2 \varphi = \rho$$

Замена в сферические координаты (ρ, φ, θ)

$$\begin{cases} x = \rho \sin \theta \cos \varphi \\ y = \rho \sin \theta \sin \varphi \\ z = \rho \cos \theta \end{cases} \quad J = \begin{vmatrix} x'_\rho & x'_\varphi & x'_\theta \\ y'_\rho & y'_\varphi & y'_\theta \\ z'_\rho & z'_\varphi & z'_\theta \end{vmatrix} = \begin{vmatrix} \sin \theta \cos \varphi & -\rho \sin \theta \sin \varphi & \rho \cos \theta \cos \varphi \\ \sin \theta \sin \varphi & \rho \sin \theta \cos \varphi & \rho \cos \theta \sin \varphi \\ \cos \theta & 0 & -\rho \sin \theta \end{vmatrix} =$$

1. $x^2 + y^2 = 4$, $x^2 + y^2 - 4x = 0$, $x = 2$

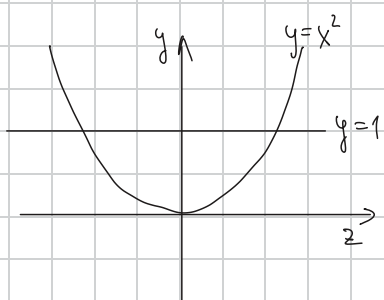


$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{cases} \quad x^2 + y^2 = \rho^2$$

$$S = \iint_D dx dy = \iint_D \rho d\rho d\varphi = \int_0^{\pi/2} d\varphi \int_2^{4 \cos \varphi} \rho d\rho = \int_0^{\pi/2} d\varphi \left[\frac{\rho^2}{2} \right]_2^{4 \cos \varphi} = \int_0^{\pi/2} (8 \cos^2 \varphi - 2) d\varphi = 4 \int_0^{\pi/2} (2 \cos^2 \varphi - 1) d\varphi = 4 \int_0^{\pi/2} \cos 2\varphi d\varphi = 2 \sin 2\varphi \Big|_0^{\pi/2} = 0$$

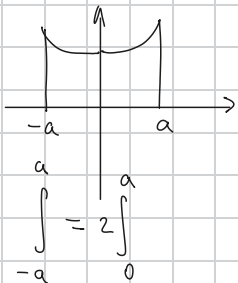
2. 3577. Параболоидом $z = x^2 + y^2$, цилиндром $y = x^2$ и плоскостями $y = 1$ и $z = 0$.

$$\iiint_{\Omega} dx dy dz = \int_{-1}^1 dx \int_{x^2}^1 dy \int_0^{x^2+y^2} dz$$



$$1) \int_0^{x^2+y^2} dz = z \Big|_0^{x^2+y^2} = x^2 + y^2$$

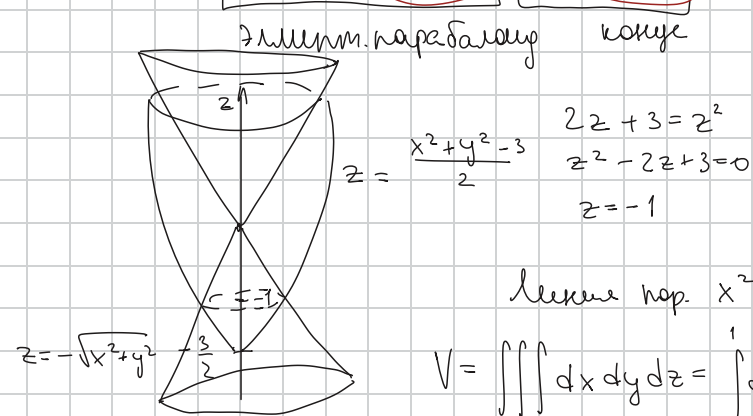
$$2) \int_{x^2}^1 (x^2 + y^2) dy = \left(x^2 y + \frac{1}{3} y^3 \right) \Big|_{y=x^2}^{y=1} = x^2 + \frac{1}{3} - x^4 - \frac{1}{3} x^6$$



$$3) \int_{-1}^1 \left(x^2 + \frac{1}{3} - x^4 - \frac{1}{3} x^6 \right) dx = 2 \int_0^1 \left(x^2 + \frac{1}{3} - x^4 - \frac{1}{3} x^6 \right) dx = 2 \left(\frac{1}{3} x^3 + \frac{1}{3} x - \frac{1}{5} x^5 - \frac{1}{21} x^7 \right) \Big|_0^1 = 2 \left(\frac{1}{3} + \frac{1}{3} - \frac{1}{5} - \frac{1}{21} \right) = \frac{88}{105}$$

Найти объем области, ограниченной поверхностями:

1.27. $2z + 3 = x^2 + y^2$, $z^2 = x^2 + y^2$ ($z < 0$).



Лемма Нор. $x^2 + y^2 = 1$

$$V = \iiint_{\Omega} dx dy dz = \int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz$$

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{cases}$$

$$= \int_0^{2\pi} d\varphi \int_0^1 \rho d\rho \int_{-\sqrt{1-\rho^2}}^{\sqrt{1-\rho^2}} dz =$$

3691. Вычислить массу тела, ограниченного параболоидом $x^2 + y^2 = 2az$ и сферой $x^2 + y^2 + z^2 = 3a^2$ ($z > 0$), если плотность в каждой точке равна сумме квадратов координат.